Software Requirements Specification (SRS)

Online Marketplace for Handcrafted Goods

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Document Review & Approval Sign-Off

Role	Name	Affiliation / Email	Sign-Off Status
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Team Lead / Development Lead	Krrish Singla	PES1UG24AM141	Submitted by Author (27-Oct-2026)
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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) establishes the authoritative requirements baseline for the "Online Marketplace for Handcrafted Goods", developed by Team 08 for the Software Engineering Mini Project course at PES University (Fall 2026). It specifies the complete functional capabilities, non-functional quality thresholds, security constraints, external integration boundaries, and formal acceptance criteria governing Sprint 1 (27–31 Oct 2026) and Sprint 2 (2–7 Nov 2026).

1.2 Scope

The Online Marketplace for Handcrafted Goods is a web-based e-commerce platform purpose-built for independent artisans and discerning buyers. The software provides an end-to-end digital storefront enabling artisans to register, configure digital artisan profiles, create item listings with high-resolution images, specify craft categories, track stock, receive real-time low-stock alerts, fulfill orders, and export sales reports. For buyers, the system delivers responsive catalog discovery with keyword search and category filtering, persistent cart management, pre-checkout inventory validation, payment processing via tokenized gateway integration, order tracking, and downloadable server-generated PDF invoices.

Explicit Exclusions: Physical logistics, third-party courier dispatch APIs, and core banking settlement internals are strictly outside the operational boundaries of this project. In Sprint 1, payment gateway interactions are verified using a deterministic local mock service stub, while Sprint 2 connects to a sandboxed payment gateway.

1.3 Intended Audience & Document Overview

This document is structured for the following project stakeholders:

• Development Team (Team 08): Serves as the implementation contract for backend APIs, relational models, and user interfaces.

• Quality Assurance Engineers: Provides precise test conditions, verification steps, and acceptance criteria referenced in the Software Test Plan (STP).

• Course Instructor & Evaluators: Establishes grading benchmarks, architectural compliance, and milestone audit criteria against the published syllabus.

1.4 Definitions, Acronyms, and Abbreviations

Term / Acronym	Formal Definition
SRS	Software Requirements Specification: this baseline requirements specification.
SAD	Software Architecture and Design Specification: companion design deliverable.
STP	Software Test Plan: companion verification and validation deliverable.
OMH	Online Marketplace for Handcrafted goods: prefix used across all requirement IDs.
FR / NFR	Functional Requirement / Non-Functional Requirement.
RTM	Requirements Traceability Matrix: bidirectional mapping between requirements, design, and tests.

Term / Acronym	Formal Definition
JWT	JSON Web Token: signed, compact token used for stateless authentication and RBAC.
RBAC	Role-Based Access Control: permission model distinguishing Buyer, Artisan/Seller, and Admin roles.
PCI-DSS	Payment Card Industry Data Security Standard: compliance standard prohibiting raw card storage.
ACID	Atomicity, Consistency, Isolation, Durability: database transaction guarantees.
SKU	Stock Keeping Unit: unique identifier for individual inventory items.
CI/CD	Continuous Integration / Continuous Deployment: automated build, test, and release pipeline.
STRIDE	Threat modeling framework: Spoofing, Tampering, Repudiation, Information Disclosure, DoS, Elevation.
WCAG	Web Content Accessibility Guidelines: international standard for digital accessibility.

2. Overall Description

2.1 Product Perspective

The Online Marketplace for Handcrafted Goods operates as a modern, self-contained multi-tier web application. It comprises a responsive client presentation layer, an API router and middleware layer, modular domain service components, and a relational persistence engine. The system integrates outward with an external payment gateway and an SMTP email relay, maintaining a clear separation of concerns.

2.2 Major Product Functions

- **Identity & Role Management**: Registration with password hashing (bcrypt), login returning signed 24-hour JWT tokens, role authorization (Buyer, Artisan, Admin), and secure password reset.
- **Product & Catalog Management**: Artisan creation and editing of handcrafted product listings with titles, descriptions, categories, pricing, stock levels, and multi-image uploads.
- **Discovery & Search Engine**: Public catalog browsing with keyword search, category filtering, pagination, and artisan storefront profile viewing.
- **Shopping Cart & Checkout**: Persistent buyer shopping cart, dynamic quantity updates, pre-checkout inventory validation, and shipping address capture.
- **Payment Processing**: Secure tokenized checkout delegating cardholder data handling to payment gateway, with automated order status updates.
- **Order Lifecycle & Tracking**: Transactional order creation, artisan fulfillment progression (Pending -> Processing -> Shipped -> Delivered), and buyer status tracking.
- **Concurrency-Safe Inventory**: Row-level locking ensuring zero overselling during simultaneous checkout attempts, paired with automated low-stock warnings.
- **Automated PDF Invoicing**: Server-side compilation of tamper-evident PDF invoices containing order metadata, line items, taxes, totals, and buyer download links.
- **Sales Analytics & Reporting**: Artisan dashboards displaying revenue metrics, sales volume, top-selling crafts, and downloadable CSV/PDF export reports.
- **Platform Moderation**: Administrator tools to oversee user accounts, inspect order transactions, and moderate flagged product listings.

2.3 User Roles and Characteristics

User Role	Characteristics & Operational Profile
Buyer (Customer)	General consumer seeking authentic handmade items. Technical capability ranges from novice to intermediate. Expects intuitive navigation, transparent pricing, secure checkout, fast response times, and accessible PDF receipts.
Artisan (Seller)	Independent craftsperson or studio owner selling handmade creations. Requires straightforward tooling to publish products, track stock, receive low-stock alerts, print or inspect orders, and view periodic sales analytics.
Platform Admin	Systems administrator responsible for platform governance, content moderation, dispute resolution, user role maintenance, and system health monitoring.

2.4 Operating Environment

• Server Runtime: Linux (Ubuntu 22.04 LTS container) running Python 3.12 with Flask/FastAPI WSGI server.
• Database Engine: Relational database (MySQL 8.0 or PostgreSQL 15) supporting ACID transactions and row-level locks.
• Client Platform: Standard modern web browsers (Google Chrome 110+, Mozilla Firefox 110+, Microsoft Edge 110+, Safari 16+) across desktop, tablet, and mobile displays.
• Deployment Engine: Docker and Docker Compose for local development and CI/CD automated staging.

2.5 Design and Implementation Constraints

- **Relational Database Mandatory**: All persistent entities (users, products, inventory, orders, payments, invoices) must be modeled and stored within a relational database schema.
- **Server-Side PDF Generation**: Invoices must be compiled deterministically on the backend using pure-python PDF engines (ReportLab), ensuring immutability and compliance.
- **PCI-DSS Compliance Boundary**: Under no circumstances shall raw credit card primary account numbers (PAN), CVVs, or PINs touch application servers or be stored in the database.
- **Two-Sprint Timebox**: Development is strictly constrained to Sprint 1 (27–31 Oct 2026) and Sprint 2 (2–7 Nov 2026), necessitating a pragmatic, modular layered architecture.
- **Single Shared Git Repository**: All project source code, documentation deliverables, and automated test suites must reside within https://github.com/krrishsingla11/SE-LAB-TEAM-08.

2.6 Assumptions and Dependencies

1. SMTP email delivery in development and automated testing uses a sandbox service (e.g. Mailtrap) to verify message headers and templates without sending actual emails.
2. Payment processing in Sprint 1 uses a local deterministic stub responding with preconfigured success and failure states, transitioning to a sandbox gateway in Sprint 2.
3. High-resolution product images are stored on local container volumes or local object stubs with relative URLs persisted in the database.

3. External Interface Requirements

3.1 User Interfaces (UI)

The web interface features responsive HTML5/CSS3 templates with clear navigation hierarchy. Key screens include: (a) Public Marketplace Landing with hero craft showcase and category grids; (b) Product Detail View with artisan profile badges and multi-image gallery; (c) Persistent Shopping Cart drawer; (d) Checkout Flow with address validation; (e) Customer Order History with one-click PDF invoice download; (f) Artisan Dashboard showing stock alerts and sales KPIs; and (g) Administrator User & Moderation Console. Color contrast complies with WCAG 2.1 AA standards.

3.2 Hardware Interfaces

Standard client computing devices (desktop PCs, laptops, tablets, and smartphones) with network connectivity. No specialized peripherals required.

3.3 Software Interfaces & External Integrations

External Software Interface	Protocol & Integration Responsibility
Payment Gateway API	RESTful HTTPS interface for tokenization and card charge authorizations. Returns JSON status payloads with unique transaction references. Stubbed for Sprint 1.
Server-Side PDF Engine	Python ReportLab library compiling structured tabular data into compliant PDF/A documents.
SMTP Email Dispatcher	Standard SMTP relay connecting to Mailtrap (development) or SendGrid for transactional order notifications and password resets.
GitHub Actions CI/CD	Automated runner executing linting, unit test suites, integration tests, and build checks upon each pull request.

3.4 Communications Interfaces & Security Protocols

All communications between client browsers and the backend server enforce TLS 1.2 or higher over HTTPS. Plaintext HTTP connections receive an automated 301 permanent redirect to HTTPS. REST endpoints exchange JSON payloads. Stateless authorization is maintained via HTTP Authorization headers using the Bearer token scheme.

4. System Features & Functional Requirements

This section details all 20 functional requirements organized by domain module. Each requirement is testable, carries a unique identifier (OMH-F-###), defines strict acceptance criteria, and links directly to corresponding test cases in the STP.

4.1 User Registration & Authentication

Req ID	Requirement Description (The system shall...)	Priority	Stakeholder	Acceptance Criteria & Test Reference
OMH-F-001	Allow new users to register by providing a unique email, strong password, full name, and selected account role (Buyer or Artisan/Seller).	High	Buyer / Artisan	AC: Valid payload creates active DB record with bcrypt hashed password. Duplicate email returns 409 Conflict. Test Ref: TC-AUTH-01, TC-AUTH-02
OMH-F-002	Authenticate existing users via email and password credentials, returning a signed JSON Web Token (JWT) valid for exactly 24 hours.	High	All Users	AC: Correct credentials yield HTTP 200 with JWT containing user_id and role claims. Invalid credentials return HTTP 401 Unauthorized. Test Ref: TC-AUTH-03, TC-AUTH-04
OMH-F-003	Enforce Role-Based Access Control (RBAC) across all protected API routes and UI views based on claims in the verified JWT.	High	System / Security	AC: Buyers requesting /api/seller/* or /api/admin/* endpoints receive HTTP 403 Forbidden. Unauthenticated requests receive HTTP 401. Test Ref: TC-AUTH-05
OMH-F-004	Provide a secure password reset mechanism generating a cryptographically random, time-limited (15-minute expiry) token delivered via email.	Medium	Buyer / Artisan	AC: Token sent to registered email; submitting token with new password updates credentials and invalidates token. Expired token returns HTTP 400. Test Ref: TC-AUTH-06, TC-AUTH-07

4.2 Product Catalog Management

Req ID	Requirement Description (The system shall...)	Priority	Stakeholder	Acceptance Criteria & Test Reference
OMH-F-005	Allow authenticated Artisans to create handcrafted product listings with title, description, category, price, stock quantity, and up to 5 images.	High	Artisan	AC: Successful submission stores product in DB, saves image files to product_images, sets active status, returns HTTP 201 with product_id. Test Ref: TC-PROD-01, TC-PROD-02
OMH-F-006	Allow Artisans to update details (price, description, stock) or soft-delete product listings that belong exclusively to their own artisan account.	High	Artisan	AC: PUT /api/products/{id} updates values. Attempts by non-owners return HTTP 403 Forbidden. Deletion sets is_active=false. Test Ref: TC-PROD-03, TC-PROD-04
OMH-F-007	Enable all users (including unauthenticated guests) to browse active product listings with keyword search, category filtering, and pagination.	High	Buyer / Public	AC: GET /api/products returns paginated array of active crafts matching query parameters. Out-of-stock items display clear badge. Test Ref: TC-PROD-05

4.3 Shopping Cart & Pre-Checkout Workflows

Req ID	Requirement Description (The system shall...)	Priority	Stakeholder	Acceptance Criteria & Test Reference
OMH-F-008	Maintain a persistent shopping cart per authenticated Buyer, allowing items to be added, quantities incremented/decremented, and items removed.	High	Buyer	AC: Cart contents persist across browser sessions. Subtotal updates dynamically based on current item unit prices. Test Ref: TC-CART-01, TC-CART-03
OMH-F-009	Validate item quantities in the cart against real-time inventory levels prior to order submission and reject checkout if stock is insufficient.	High	Buyer / System	AC: Checkout attempt with requested quantity > available inventory returns HTTP 400 Bad Request identifying out-of-stock items. Test Ref: TC-CART-02, TC-CHK-01

4.4 Payment Processing & PCI Safety

Req ID	Requirement Description (The system shall...)	Priority	Stakeholder	Acceptance Criteria & Test Reference
OMH-F-010	Initiate payment transactions via payment gateway interface (stubbed in Sprint 1) and process authorization responses before committing orders.	High	Buyer / System	AC: Approved gateway response advances payment status to "captured", initializes order fulfillment status to "Processing", and records transaction reference. Declined response releases row locks, aborts order creation, and returns HTTP 402. Test Ref: TC-PAY-01, TC-PAY-02, TC-PAY-03
OMH-F-011	Prohibit the transmission, caching, or storage of raw cardholder numbers (PAN) or security codes (CVV) on marketplace application servers.	High	Security / Compliance	AC: Application only receives and stores opaque payment tokens and transaction references; database audit verifies zero card numbers. Test Ref: TC-SEC-02

4.5 Order Management & Tracking

Req ID	Requirement Description (The system shall...)	Priority	Stakeholder	Acceptance Criteria & Test Reference
OMH-F-012	Create an immutable Order record upon payment confirmation, capturing buyer ID, line items, historical unit prices, shipping address, and total amount.	High	Buyer / Artisan	AC: Atomic DB transaction records order and order items with fulfillment status "Processing" and payment record status "captured". Emits event for invoice compilation. Test Ref: TC-ORD-01
OMH-F-013	Allow Artisans to view incoming orders for their handcrafted items and transition fulfillment status through Pending, Processing, Shipped, and Delivered.	High	Artisan	AC: Artisan updates order status via dashboard; valid transitions persist and are visible to buyer in real time. Invalid transitions return HTTP 400. Test Ref: TC-ORD-02
OMH-F-014	Send automated email notifications to buyers upon order placement, payment confirmation, and status progression to Shipped or Delivered.	Medium	Buyer	AC: Order status change dispatches templated notification containing order ID and tracking status to buyer email address. Test Ref: TC-ORD-03

4.6 Inventory Management & Low-Stock Alerts

Req ID	Requirement Description (The system shall...)	Priority	Stakeholder	Acceptance Criteria & Test Reference
OMH-F-015	Execute atomic, concurrency-safe inventory decrements upon order placement using database row-level locking to prevent overselling.	High	System / Artisan	AC: Concurrent purchases for the last remaining unit result in exactly one order success and one out-of-stock failure; stock never falls below 0. Test Ref: TC-INV-01
OMH-F-016	Detect when product inventory falls below the artisan-configured threshold (default: 5 units) and trigger an automated low-stock dashboard alert.	Medium	Artisan	AC: Inventory update or sale causing stock <= threshold adds visual alert badge and entry to artisan dashboard notification queue. Test Ref: TC-INV-02

4.7 Server-Side PDF Invoice Generation & Retrieval

Req ID	Requirement Description (The system shall...)	Priority	Stakeholder	Acceptance Criteria & Test Reference
OMH-F-017	Generate a formal, tamper-evident PDF invoice for every successfully processed order (status "Processing") containing invoice number, order date, buyer details, itemized lines, and tax totals.	High	Buyer / Compliance	AC: Completed order automatically compiles PDF using ReportLab; PDF file is saved to secure storage and linked to order record. Test Ref: TC-INV-03
OMH-F-018	Provide authenticated Buyers and Artisans with direct HTTP download endpoints to retrieve PDF invoices corresponding to their own orders.	Medium	Buyer / Artisan	AC: GET /api/orders/{id}/invoice streams PDF with Content-Type: application/pdf. Access attempts for orders not owned return HTTP 403. Test Ref: TC-INV-04

4.8 Sales Analytics & Report Export Workflows

Req ID	Requirement Description (The system shall...)	Priority	Stakeholder	Acceptance Criteria & Test Reference
OMH-F-019	Generate aggregated sales analytics for authenticated Artisans, displaying gross revenue, units sold, order count, and top craft items over selected date ranges.	High	Artisan	AC: GET /api/reports/sales?period=monthly computes accurate aggregate figures derived from completed orders in the database. Test Ref: TC-RPT-01
OMH-F-020	Enable Artisans to export sales summary reports and current inventory stock status as formatted CSV spreadsheets or downloadable PDF summaries.	Medium	Artisan	AC: GET /api/reports/export?format=csv returns properly formatted CSV stream; format=pdf returns compiled summary document. Test Ref: TC-RPT-02

5. Non-Functional Requirements (NFRs)

5.1 Measurable Quality Attributes

Req ID	Requirement Statement	Quality Category	Priority	Measurement Metric & Acceptance Criterion
OMH-NF-001	The system shall respond to at least 95% of standard read and search API requests within 500 milliseconds under a nominal load of 100 concurrent users.	Performance	High	JMeter load test: 95th percentile response latency <= 500 ms; 0 HTTP 5xx errors. Test Ref: TC-PERF-01
OMH-NF-002	The marketplace web service and database shall maintain at least 99.5% monthly availability, excluding scheduled maintenance windows of <= 2 hours/week.	Reliability	High	Automated health-check monitor reports >= 99.5% successful pings over monthly operational window. Test Ref: TC-OPS-01
OMH-NF-003	All data in transit across public networks shall be encrypted using TLS 1.2 or higher; plaintext HTTP connections shall be immediately rejected.	Security	High	SSL Labs configuration grade >= A; plaintext requests receive 301 redirect. Test Ref: TC-SEC-01
OMH-NF-004	The stateless API tier and relational database connection pool shall support at least 500 concurrent active sessions without data corruption or crashes.	Scalability	Medium	JMeter concurrency stress test with 500 virtual users executing simultaneous browsing and cart operations. Test Ref: TC-PERF-02
OMH-NF-005	The user interface shall be fully responsive across desktop and mobile browsers, satisfying WCAG 2.1 AA accessibility guidelines.	Usability / A11y	Medium	Automated axe-core accessibility audit reports 0 critical or high violations; keyboard navigability verified. Test Ref: TC-UX-01

5.2 Security Objectives

- **SO-1: Data Confidentiality & Credential Protection**: Ensure all user credentials, personal contact data, and financial transactions are protected against unauthorized access, eavesdropping, and exfiltration both in transit and at rest.
- **SO-2: Transactional Integrity & Fraud Prevention**: Guarantee that product pricing, inventory deductions, order totals, and payment records cannot be tampered with or repudiated by unauthorized actors or malicious client payloads.

5.3 Security Requirements

Req ID	Security Requirement Statement	Category	Priority	Acceptance Criterion & Test Ref
OMH-SR-001	User passwords shall be hashed using bcrypt with a work factor of at least 12 prior to database storage; plaintext passwords shall never be logged or persisted.	Security	High	DB inspection verifies all stored hashes adhere to $2b$12$ format; no plaintext in logs. Test Ref: TC-SEC-03
OMH-SR-002	Stateless JWT tokens shall enforce cryptographic signatures (HMAC-SHA256), a 24-hour expiration window, and role claim validation on every protected route.	Security	High	Expired or tampered token presented to protected endpoint returns HTTP 401 Unauthorized. Test Ref: TC-SEC-04

Req ID	Security Requirement Statement	Category	Priority	Acceptance Criterion & Test Ref
OMH-SR-003	All user-supplied input parameters shall be validated against strict schemas and sanitised using parameterised ORM queries to eliminate SQL injection and XSS.	Security	High	OWASP ZAP vulnerability scan yields 0 high-risk SQLi or reflected/stored XSS findings. Test Ref: TC-SEC-05
OMH-SR-004	The authentication service shall enforce IP-based rate limiting, blocking further login attempts after 5 consecutive failures within a 10-minute window.	Security	High	Sixth consecutive failed login from an identical IP address returns HTTP 429 Too Many Requests. Test Ref: TC-SEC-06
OMH-SR-005	The system shall maintain structured, immutable audit logs recording all authentication events, financial transactions, and order state mutations.	Security	High	Audit log entries contain UTC timestamps, user IDs, IP addresses, and event outcomes; zero sensitive data logged. Test Ref: TC-SEC-07

6. Quality Attributes & Acceptance Tests

To achieve formal sign-off for Sprint 1, Sprint 2, and the Final Project Review, the implementation must satisfy the following exit criteria:

1. 100% of all High-priority functional requirements (OMH-F-001 through OMH-F-020) implemented and verified.
2. Zero open Critical or High severity defects logged in the defect tracking system.
3. All performance, scalability, and security thresholds defined in Section 5 verified through automated test suites.
4. Bidirectional traceability verified via the Requirements Traceability Matrix (RTM).
5. Working end-to-end demonstrations recorded and submitted per course milestones (Sprint 1: 1-minute video; Sprint 2: 2-minute video).

7. UML Use-Case Models: 7.1 Buyer End-to-End Shopping Experience

Figure 1 depicts the primary interactions initiated by the Buyer actor and the external Payment Gateway. Checkout inherently «includes» stock validation and payment processing, while search filters «extend» catalog browsing.

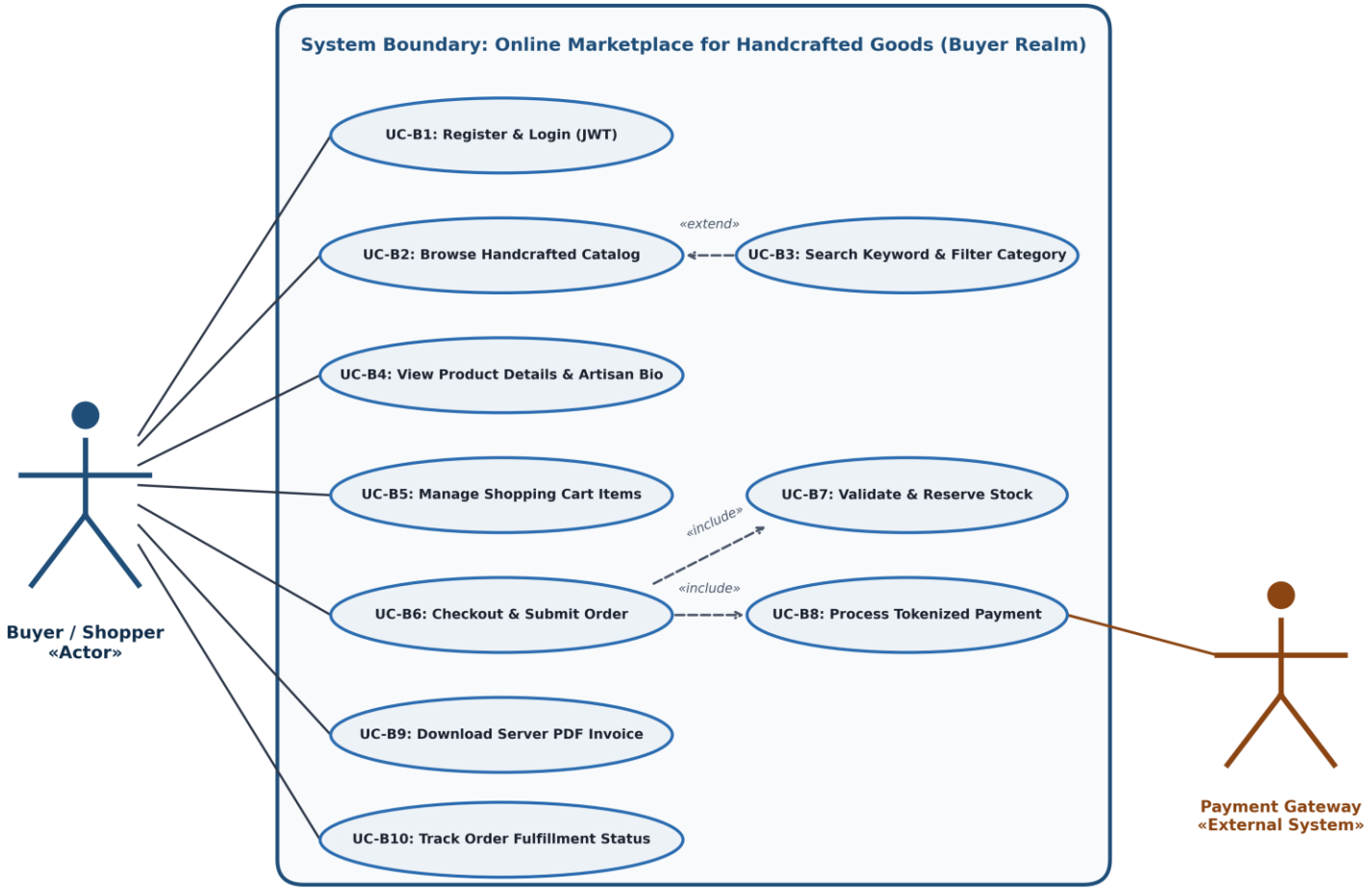


Figure 1: UML Use-Case Diagram – Buyer Shopping Experience & Checkout Workflows

7.2 Use-Case Model 2: Artisan Storefront & Administrative Management

Figure 2 illustrates independent operational workflows executed by the Artisan/Seller actor and Platform Admin actor. Artisans govern product creation, detail editing, deactivations, stock monitoring, fulfillment, and report exports, while Administrators oversee moderation, role governance, and platform auditing.

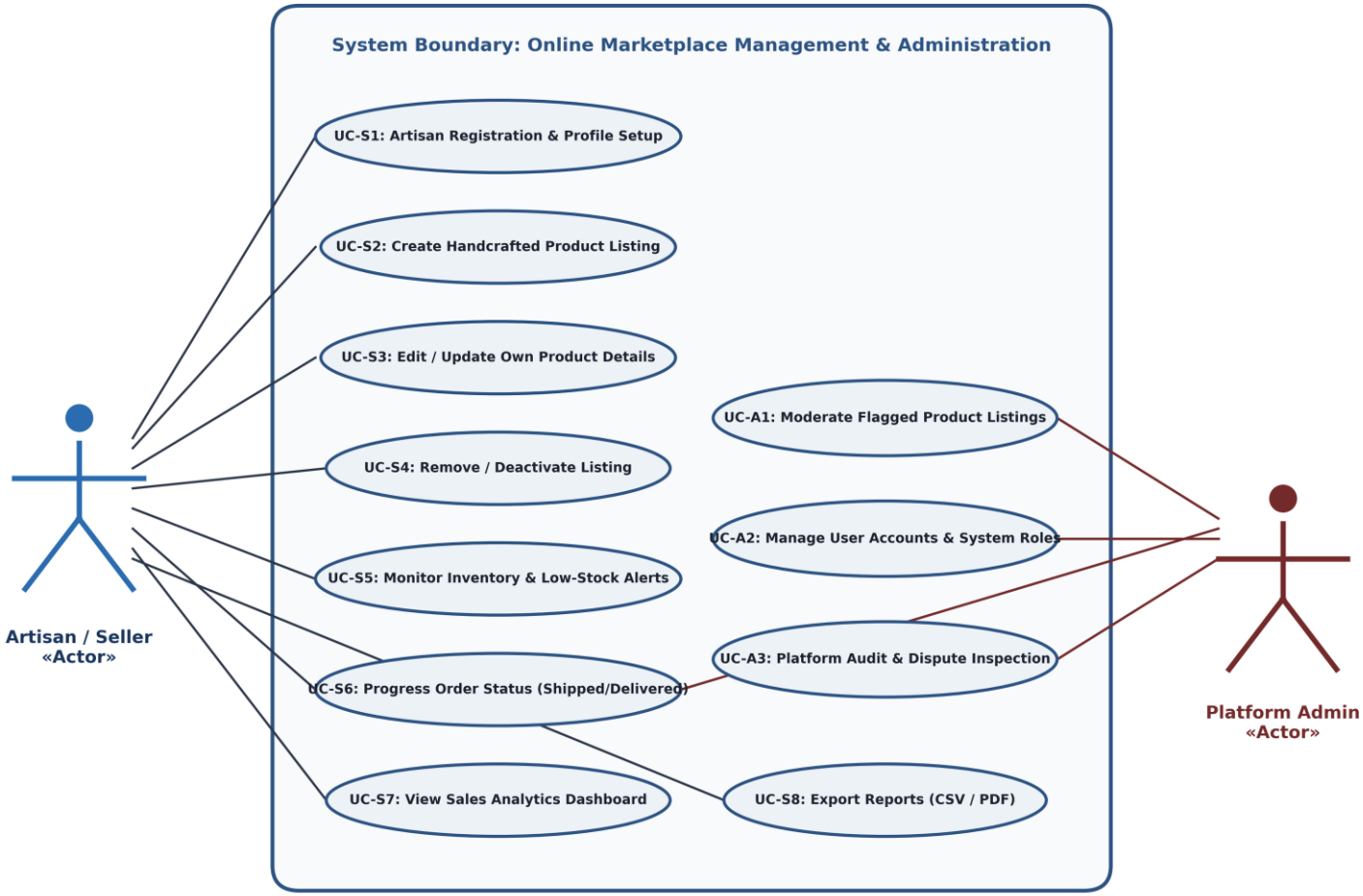


Figure 2: UML Use-Case Diagram – Artisan Storefront Operations & System Administration

8. Requirements Traceability Matrix (RTM)

The RTM establishes forward and backward traceability connecting each SRS requirement to its architectural design module in the SAD, corresponding verification test cases in the STP, implementation sprint, and planned verification status.

Req ID	Requirement Summary	SAD Module & Design Ref	STP Test Case(s)	Sprint	Status
OMH-F-001	User Registration	AuthService / DS-AUTH-01	TC-AUTH-01, TC-AUTH-02	1	Planned
OMH-F-002	User Login (JWT)	AuthService / DS-AUTH-02	TC-AUTH-03, TC-AUTH-04	1	Planned
OMH-F-003	RBAC Enforcement	JWT Middleware / DS-AUTH-03	TC-AUTH-05	1	Planned
OMH-F-004	Password Reset Token	AuthService / DS-AUTH-04	TC-AUTH-06, TC-AUTH-07	1	Planned
OMH-F-005	Create Product Listing	CatalogService / DS-CAT-01	TC-PROD-01, TC-PROD-02	1	Planned
OMH-F-006	Update/Delete Listing	CatalogService / DS-CAT-02	TC-PROD-03, TC-PROD-04	1	Planned
OMH-F-007	Browse & Search Catalog	CatalogService / DS-CAT-03	TC-PROD-05	1	Planned
OMH-F-008	Persistent Shopping Cart	CartService / DS-CART-01	TC-CART-01, TC-CART-03	1	Planned
OMH-F-009	Pre-Checkout Stock Check	CartService / DS-CART-02	TC-CART-02, TC-CHK-01	1	Planned
OMH-F-010	Payment Processing (Stub)	PaymentService / DS-PAY-01	TC-PAY-01, TC-PAY-02, TC-PAY-03	1/2	Planned
OMH-F-011	Zero Cardholder Storage	PaymentService / DS-PAY-02	TC-SEC-02	1/2	Planned
OMH-F-012	Transactional Order Create	OrderService / DS-ORD-01	TC-ORD-01	2	Planned
OMH-F-013	Artisan Order Fulfillment	OrderService / DS-ORD-02	TC-ORD-02	2	Planned
OMH-F-014	Order Event Notifications	OrderService / DS-ORD-03	TC-ORD-03	2	Planned
OMH-F-015	Atomic Stock Decrement	InventoryService / DS-INV-01	TC-INV-01	2	Planned
OMH-F-016	Low-Stock Alert Trigger	InventoryService / DS-INV-02	TC-INV-02	2	Planned
OMH-F-017	Server PDF Invoice Creation	InvoiceService / DS-INV-03	TC-INV-03	2	Planned
OMH-F-018	Customer Invoice Download	InvoiceService / DS-INV-04	TC-INV-04	2	Planned
OMH-F-019	Sales Analytics Dashboard	ReportService / DS-RPT-01	TC-RPT-01	2	Planned
OMH-F-020	Export Reports (CSV/PDF)	ReportService / DS-RPT-02	TC-RPT-02	2	Planned
OMH-NF-001	p95 Response <= 500 ms	Web Layer / DS-PERF-01	TC-PERF-01	2	Planned
OMH-NF-002	99.5% Service Availability	Infrastructure / DS-REL-01	TC-OPS-01	2	Planned
OMH-NF-003	Mandatory TLS 1.2+ HTTPS	Gateway / DS-SEC-01	TC-SEC-01	1	Planned
OMH-NF-004	500 Concurrent Sessions	API / DB Pool / DS-PERF-02	TC-PERF-02	2	Planned
OMH-NF-005	WCAG 2.1 AA Usability	Web Client / DS-UX-01	TC-UX-01	2	Planned
OMH-SR-001	bcrypt Password Hashing	AuthService / DS-SEC-02	TC-SEC-03	1	Planned
OMH-SR-002	JWT 24-Hour Expiry Check	JWT Middleware / DS-SEC-03	TC-SEC-04	1	Planned
OMH-SR-003	Input Sanitisation & SQLi	ORM / Sanitizer / DS-SEC-04	TC-SEC-05	2	Planned
OMH-SR-004	Login Rate Limiter (5/10m)	RateLimiter / DS-SEC-05	TC-SEC-06	2	Planned
OMH-SR-005	Immutable Security Audit Log	LoggingService / DS-SEC-06	TC-SEC-07	2	Planned